

2	(a)	speed decreases/stone decelerates <u>to rest/zero</u> at 1.25 s	B1	
		speed then increases/stone accelerates (in opposite direction)	B1	[2]
	(b)	(i) $v = u + at$ (or $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$)	C1	
		$= 0 + (3.00 - 1.25) \times 9.81$	C1	
		$= 17.2 \text{ (17.17) ms}^{-1}$	A1	[3]
		(ii) $s = ut + \frac{1}{2}at^2$		
		$s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$	C1	
		$s = \frac{1}{2} \times 9.81 \times (1.75)^2 [= 15.02]$	C1	
		(distance = 7.66 + 15.02)		
		$[v = u + at = 0 + 9.81 \times (2.50 - 1.25) = 12.26 \text{ ms}^{-1}]$		
		or		
		$s = \frac{1}{2} \times 9.81 \times (1.25)^2 [= 7.66]$	(C1)	
		$s = 12.26 \times 0.50 + \frac{1}{2} \times 9.81 \times (3.00 - 2.50)^2 [= 7.36]$	(C1)	
		(distance = $2 \times 7.66 + 7.36$)		
		<i>Example alternative method:</i>		
		$s = (v^2 - u^2)/2a = (12.26^2 - 0)/2 \times 9.81 [= 7.66]$	(C1)	
		$s = (v^2 - u^2)/2a = (17.17^2 - 12.26^2)/2 \times 9.81 [= 7.36]$	(C1)	
		(distance = $2 \times 7.66 + 7.36$)		

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22.7 (22.69 or 23) m

A1 [3]

(iii) ($s = 15.02 - 7.66 =$) 7.4 (7.36) m (*ignore sign in answer*)

A1

down

A1 [2]

(c) straight line from positive value of v to t axis

M1

same straight line crosses t axis at $t = 1.25$ s

A1

same straight line continues with same gradient to $t = 3.0$ s

A1 [3]